

A Cortico-Cerebellar model for non-markovian tasks

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What is the actual problem?

- ▶ With reversal-learning task lets see if we can derive a value and uncertainty signal
- ▶ What, in more general terms, is a reversal-learning task?
- ▶ If option $A \rightarrow +0$
- ▶ Does the (Action, Reward) tuple at time t provide enough information to determine if option $A < \text{option } B$
- ▶ Agent must know the last N trials to determine $A > B \mid A < B$
- ▶ Agent wants to determine (Reward $\mid \pi$), so it must derive $\text{Value}(\pi) \mid f(O_t, O_{t-1}, \dots, O_{t-N})$, where O_t is the observed reward and environment at time t
- ▶ This gives us a solid frame-work to start with, formally this same problem is partially observable markov decision process (POMDP)

Reversal-learning is a non-markovian environment for a limited information agent

- ▶ We know that in a POMDP there's a state S_t which transition to S_{t+1} is markovian, thus depends only on S_t
- ▶ However, we also know that from the perspective of a limited information agent, the environment behaves non-markovian as $O_t \rightarrow O_{t+1}$ depends on S which is unknown and presents as a latent state
- ▶ Ideal agent needs to derive $S = f(O_t, \dots)$, in other words, the agent needs to know if its winter or autumn by checking if its raining or not

TLDR

- ▶ Reversal learning $\rightarrow$ non-markovian
- ▶ non-markovian $\rightarrow$ requires temporal integration
- ▶ The integration of $S, R, V(\pi) \rightarrow$ Eligibility trace

Solving non-markovian environments

- ▶ Resting-state fMRI (Habas et al., 2009) → whole-brain ICA → template matching → functional template (Damoiseaux et al., 2006)
- ▶ ↑ Executive control network (Crus I, II)
- ▶ ↑ Salience network (DCN)
- ▶ We could speculate a continuous injection of historical states/errors
- ▶ More evidence by Buckner et al., 2011, showing association networks >> motor networks

Optogenetic evidence in non-markovian environments

(Gao et al., 2018)

- ▶ Anterior lateral motor cortex (ALM, activity predicting future licking)
- ▶ DCN perturbation/lesion $\rightarrow$ $\downarrow$ non-markovian task performance
- ▶ No changes in response stage
- ▶ DCN $\rightarrow$ VM-thalamus $\rightarrow$ ALM
- ▶ Dentate acts as control

Specificity of function in non-markovian environments

(Gao et al., 2018; Kalmbach et al., 2009)

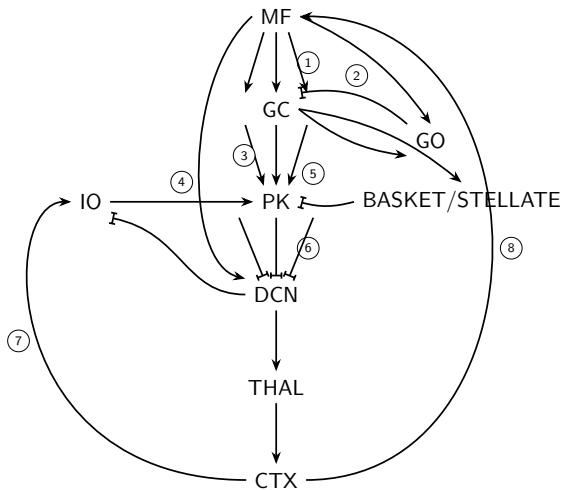
Perturbed Region	Pathway Severed	Markovian	Non-Markovian
		<i>(Delay / Cue-Guided)</i>	<i>(Trace / Delayed-Response)</i>
None	Intact	OK	OK
DCN	Cb → Ctx	FAIL	FAIL
mPFC	Ctx → Cb	OK	FAIL
DCN → Thal	Cb → Ctx	OK	FAIL

Policy perserveration in non-markovian tasks

(Verpeut et al., 2023) Fig 2B, 2C

- ▶ Chemogenetic perturbation of cerebellum impairs reversal-learning → perseverance
- ▶ Mice learn simple markovian pairing during training
- ▶ Mice show perserveration despite no reward
- ▶ Correct answer over session are almost purely at second try, without decreasing it. Thus, the implement a lose-shift heuristic instead of learning the new rule

Cerebellum canonical model



- ① Spatio-temporal expansion
- ② Sparsity regulation
- ③ Eligibility trace (*Buffer state forward*)
- ④ Teaching signal

- ⑤ Long term depression (*Anti-Hebbian plasticity*)
- ⑥ Inhibitory compression (*Authors didn't think of this!*)
- ⑦ Cortical prediction error
- ⑧ Efference copy

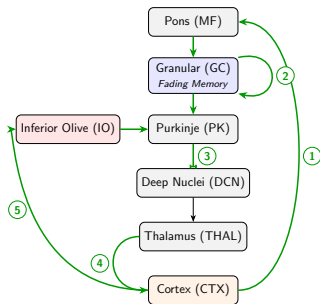
The gap in canonical models

- ▶ **Fading memory.** how does the granule layer buffer historical states across non-markovian temporal (or other) gaps
- ▶ **Cortico-Cerebellar feedback loop.** canonical models heavily favored a *feedforward* mode in the cerebellum (though as pure motor controller). Largely ignoring or not explicitly modeling the projections from DCN to the CTX via Thal
- ▶ **Credit assignment.** canonical models relied on instantaneous motor errors. Lacking a mechanism to map a delayed, abstract reward signal back onto a sequence of past hidden states, to overcome cognitive policy perserveration

The Goal of Our Model

To mathematically formulate and simulate these missing recurrent and temporal dynamics, proving the cerebellum's capacity to resolve non-Markovian environments. Then if we capture a value and uncertainty signal it would be a more 'emergent' less bias finding.

Bridging the Gap



1. **State Expansion (μ_t):** The cortical state is projected to the cerebellum via the mossy fibers.
2. **The Fading Memory (G_t):** The granular layer is modeled as a recurrent Echo State Network (ESN), generating the temporal eligibility trace to bridge the gap.
3. **State Compression:** The Purkinje/DCN bottleneck compresses the high-dimensional historical trace into $\mathbf{D}_t$.
4. **Thalamic Feedback (β_{thal}):** Continuous injection of $\mathbf{D}_t$ back into the cortex, transforming the system from a Markovian actor into a Non-Markovian one.
5. **Historical Credit Assignment:** The cortical prediction error ϵ_t drives anti-Hebbian plasticity at the GC $\rightarrow$ PK synapses to update the global policy.

Cortex hallucination of sensory input

$$\hat{\mathbf{I}}_t = \mathbf{W}_{gen}\boldsymbol{\mu}_t \quad (1)$$

- ▶ $\hat{\mathbf{I}}$, is the *hallucination* of equal dimension to $\mathbf{I}$ *actual input*
- ▶ $\mathbf{W}_{gen}$ is the generative matrix than makes the lower-dimension projection, intialized random and fixed $\mathcal{N}(0, 1/\sqrt{\dim \boldsymbol{\mu}})$
- ▶ $\boldsymbol{\mu}_t$ is the *cortical state* at time t

Cortical prediction error

$$\epsilon_t = \Pi \odot (\mathbf{I}_t - \hat{\mathbf{I}}_t) \quad (2)$$

- ▶ The cortex makes a *hallucination* which acts as active inference of the world, then the participant makes an actual choice generating the sensory vector $\mathbf{I}_t$
- ▶ To build Π as the precision weighting, we choose the most simplest one. Static anatomical precision, let $\Pi_{raw,i} = 1/\sigma(\mathbf{I}_i)$ be the raw precision weighting vector for input i with variance $\sigma(\mathbf{I}_i)$, then we generate $\Pi = \Pi_{raw}/\bar{\Pi}_{raw}$ to be the normalized precision weighting
- ▶ Element-wise multiplication means we build prediction error ϵ_t as the precision weighted prediction error

Propagation of cortical state in to the Cerebellum

$$\mathbf{G}_t = \mathbf{W}_{ach1} \boldsymbol{\mu}_t \quad (3)$$

- ▶ $\mathbf{G}_t$ is the Granule Layer state at time t , defined as a Marr-Albus-Ito random sparse projection
- ▶ We modeled this as an Achlioptas random sparse projection (896×32), where each weight $w_{i,j}$ is assigned with probability p : -1 or 1 , and 0 with $1 - 2p$, setting sparsity ($1 - 2p$) to 90% zeros

Cerebellum memory or eligibility trace

When the cortex makes a prediction and initiates an action, there is some delay for this feedback to arrive. How do we solve the problem of pairing some particular $\mathbf{G}_t$ with a given feedback. We need to leave an *eligibility trace*

$$\mathbf{Z}_t = (1 - \beta_{gran})\mathbf{Z}_{t-1} + \alpha_{gran}\mathbf{G}_t \quad (4)$$

- ▶ Where $\mathbf{G}_t$ is the instantaneous firing of the Granule Layer
- ▶ α_{gran} is the parameter that controls how aggressively the layer updates
- ▶ $(1 - \beta_{gran})$ is the *leaky bucket*, that decays the trace on every trial

Cerebellum learning

To learn the Cerebellum projects the 6-Dimensional precision weighted error ϵ_t to a single dimension (relating this to the Inferior Olive)

$$|\bar{\epsilon}_t| = \frac{1}{d} \sum_{i=1}^d |\epsilon_{t,i}| \quad (5)$$

With that scalar error we update the Granule Layer trace in its Purkinje Cell representation via anti-Hebbian learning

$$\mathbf{W}_{purk,t} = \mathbf{W}_{purk,t-1} - \kappa(|\bar{\epsilon}_t| \mathbf{Z}_t) \quad (6)$$

Cerebellum compression

$$\mathbf{D}_t = \mathbf{W}_{ach2}(\mathbf{G}_t \odot \mathbf{W}_{purk,t}) \quad (7)$$

- ▶ $\mathbf{D}_t \in \mathbb{R}^{362}$ is the Cerebellum forward pass
- ▶ $\mathbf{G}_t \odot \mathbf{W}_{purk,t}$ is the element-wise multiplication between the Granule Layer current state and the LTD Purkinje layer we previously constructed
- ▶ Finally with $\mathbf{W}_{purk,t}$ we perform a random sparse matrix compression into the 362 dimension of the cerebellar forward pass

The loop-back: thalamo-cortical pass

$$\boldsymbol{\mu}_{t+1} = \boldsymbol{\mu}_t + \alpha_c \left[(\mathbf{W}_{gen}^T \boldsymbol{\epsilon}_t) - (\lambda_c \cdot ITI_t \cdot \boldsymbol{\mu}_t) + \beta_{thal}(\mathbf{W}_{thal} \mathbf{D}_t) \right] \quad (8)$$

- ▶ This is the Cortical trace update, $\mathbf{W}_{gen}^T \boldsymbol{\epsilon}_t$ projects the error back into the cortical dimension $6 \rightarrow 32$, and we perform gradient descent over a time-decaying trace $\lambda_c \cdot ITI_t \cdot \boldsymbol{\mu}_t$
- ▶ $\mathbf{W}_{thal}$ is the connectivity from cerebellum to cortex $362 \rightarrow 32$, so $\mathbf{W}_{thal} \cdot \mathbf{D}_t$ is the cerebellum-cortex forward pass, which via β_{thal} pushes the cortex into the non-linear projection
- ▶ Finally, the trace is updated via learning rate α_c

Converting a trace into action and reaction-time

For empirical prediction, M_{cc} utilizes an L2-regularized Ridge regression readout to map the cortical latent state $\boldsymbol{\mu}_t$ onto the observed behavioral variables (binary choice and continuous reaction time):

$$P(\text{Choice}_t = 1 \mid \boldsymbol{\mu}_t) = \sigma(\mathbf{w}_{\text{choice}}^T \boldsymbol{\mu}_t) \quad (9)$$

$$RT_t \sim \mathcal{N}(\mathbf{w}_{\text{rt}}^T \boldsymbol{\mu}_t, \sigma_{\text{rt}}^2) \quad (10)$$

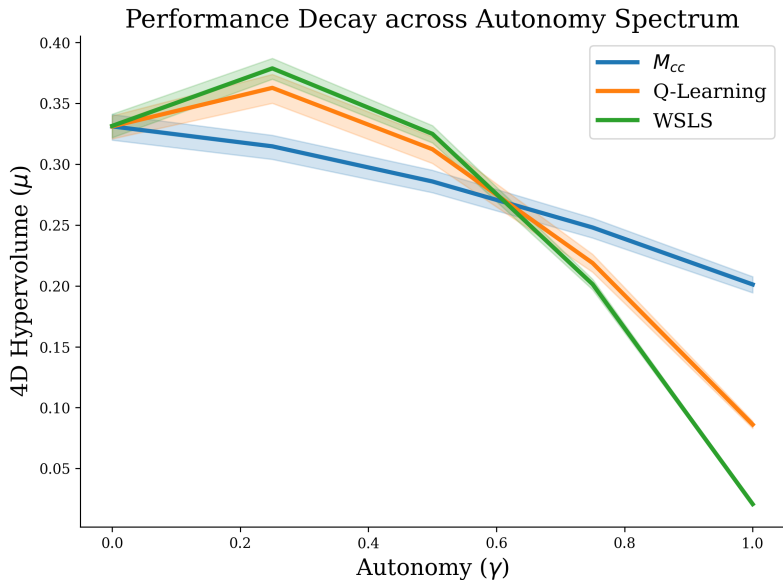
where $\sigma(\cdot)$ denotes the logistic sigmoid function, and $\mathbf{w}_{\text{choice}}$ and $\mathbf{w}_{\text{rt}}$ are the respective optimal linear readout weights.

Methods: Teacher-Student Distillation

- ▶ **The Teacher Proxy ($RNN_{teacher}$):** We trained a Gated Recurrent Unit (GRU) to predict both a binary choice policy (BCE) and a continuous reaction time kinematic head (Log-Normal mixture).
- ▶ **Calibration:** The RNN's raw logits were Platt-scaled to produce true probability distributions, scored via Continuous Ranked Probability Score (CRPS).
- ▶ **The Distillation Loss ($\mathcal{L}_{Distill}$):** M_{cc} continuously projected its cortical state (μ_t^{bio}) to match the Teacher's latent state (μ_t^{RNN}). This was formalized as a robust Gaussian likelihood over the residual space:

$$\mathcal{L}_{Distill} = \sum_t \left[-\frac{1}{2} \ln(2\pi\sigma_{res}^2) - \frac{(\mu_t^{RNN} - \mathbf{W}_{proj}\mu_t^{bio})^2}{2\sigma_{res}^2} \right] \quad (11)$$

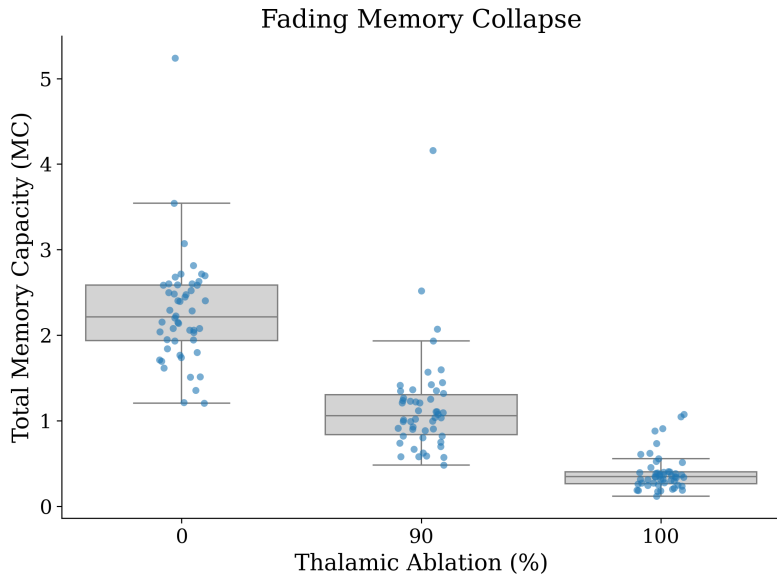
Results: NSGA-II Parameter Discovery



Results: NSGA-II Parameter Discovery (Statistics)

- ▶ **Multi-Objective Front:** We searched the structural space using NSGA-II across a 5D objective space, jointly minimizing distillation loss and empirical behavioral loss, while explicitly penalizing size with an $O(N^{2.5})$ metabolic cost.
- ▶ **Optimal Topology:** Projecting the 5D solutions back to a 4D performance space converged to strict structural limits: 37.5% of Granular capacity (≈ 336 cells) and 18.4% of DCN capacity (≈ 165 cells).
- ▶ **Hypervolume Decay:** Across the autonomy spectrum ($\gamma \in [0, 1]$), M_{cc} degraded significantly slower ($\beta_{log} = -0.494$, 95% CI $[-0.504, -0.482]$) than Q-Learning ($\beta_{log} = -1.279$, $p < .001$) and WSLS ($\beta_{log} = -2.464$, $p < .001$).

Results: Fading Memory & Thalamic Ablation



Results: Fading Memory & Thalamic Ablation (Statistics)

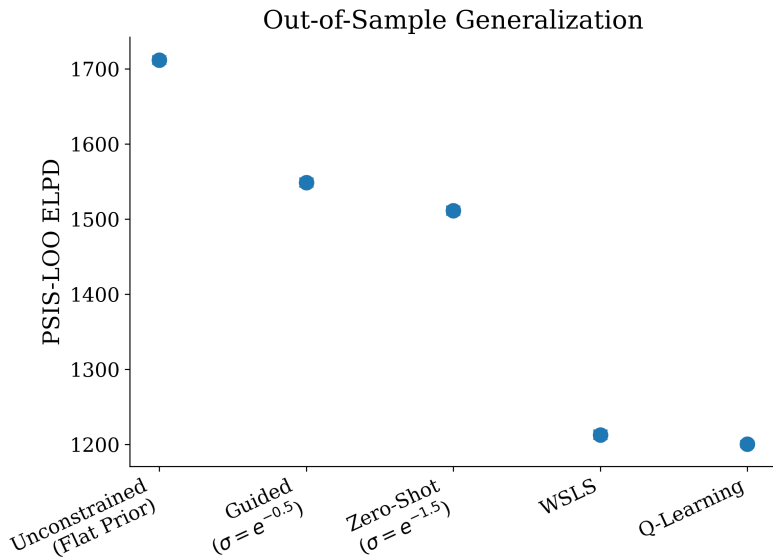
- ▶ **Non-Markovian Integration:** To verify temporal buffering independent of the Teacher, we evaluated Memory Capacity (MC) up to $k = 15$ past lags using an L2-Ridge readout on the cortical space.
- ▶ **Intact Capacity:** The baseline M_{cc} maintains significant temporal memory ($TotalMC = 0.809$) compared to a temporally permuted null baseline.
- ▶ **Ablation Collapse:** Sequential synthetic ablation of the thalamic tract (β_{thal}) induced a massive, linear collapse in temporal buffering (Estimate = -1.327, $t(49) = -30.34$, $p < .0001$). At 100% complete lesion, the network collapsed into a purely instantaneous Markovian reactor (Total MC = 0.065).

Methods: Bayesian Inference & Generalization (Math & Intuition)

- ▶ **Adaptive Metropolis-Hastings (MCMC):** To fit the hierarchical behavioral models, we utilized a custom C++ Adaptive Random Walk Metropolis-Hastings algorithm (Haario et al., 2001). **Intuition:** The sampler iteratively proposes parameter updates using an empirically adapted covariance matrix, dynamically scaling the step size via a Robbins-Monro stochastic approximation to maintain an optimal 23.4% acceptance rate.
- ▶ **Leave-One-Out Cross Validation:** We evaluated out-of-sample generalization using Pareto-Smoothed Importance Sampling (PSIS-LOO) (Vehtari et al., 2017).
- ▶ **Expected Log Predictive Density (ELPD):** PSIS-LOO estimates how well the model predicts held-out data.

$$\text{ELPD} = \sum_{i=1}^N \log \int p(y_i | \theta) p(\theta | y_{-i}) d\theta \quad (12)$$

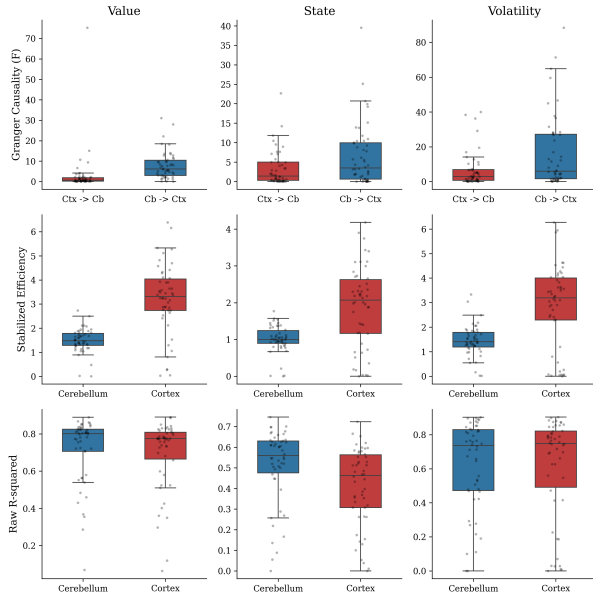
Results: Bayesian Factorial Design & PSIS-LOO



Results: Bayesian Factorial Design & PSIS-LOO (Statistics)

- ▶ **Zero-Shot Generalization:** Using 10,000 MCMC iterations per subject, we computed out-of-sample ELPD via PSIS-LOO.
- ▶ **Unconstrained Overfitting:** A flat prior ($\mathcal{N}(0, 1)$) yielded an ELPD of 1712.0 but completely failed validation, throwing 389 Pareto $k > 0.7$ warnings due to unchecked posterior variance.
- ▶ **Physiological Distillation:** Using the NSGA-II generated guided priors ($\sigma = e^{-0.5}$) completely bounded the geometry (0 warnings, ELPD = 1548.6). This architecture significantly overpowered the WSLS ($\Delta\text{ELPD} = 335.9, p < .001$) and Q-Learning heuristics, proving the loop strictly generalizes behavioral temporal dynamics.

Results: Directional Flow & Raw Encoding



Methods: Quantifying Flow & Encoding Capacity (Math & Intuition)

- ▶ **Granger Causality (F):** Does the past temporal sequence of the Cerebellum's state help predict the future of the Cortex's state better than the Cortex's own past alone? Higher F -values indicate stronger directional information flow.
- ▶ **Raw Encoding Capacity (R^2):** L2-regularized Ridge regression to decode the latent cognitive variable (y_t) solely from the region's continuous state vector ($\mathbf{X}_t$).

$$R^2 = 1 - \frac{\sum (y_t - \mathbf{X}_t \mathbf{w})^2}{\sum (y_t - \bar{y})^2} \quad (13)$$

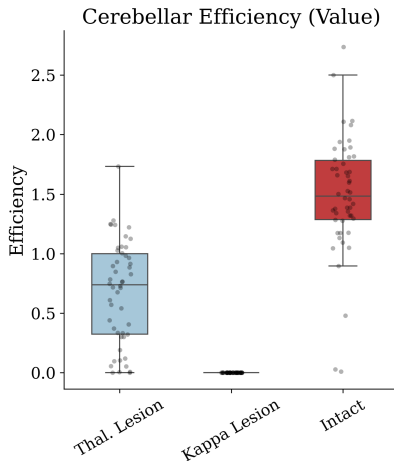
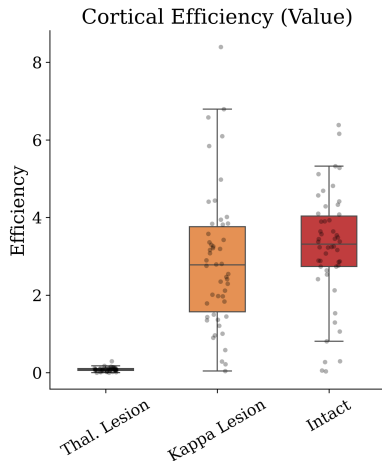
- ▶ **Stabilized Readout Efficiency:** We penalized the raw predictive power by the intrinsic temporal volatility (trajectory derivative) of the region's state space:

$$\text{Efficiency} = \frac{R^2}{\text{Volatility}} = \frac{R^2}{\mathbb{E}[||\mathbf{X}_t - \mathbf{X}_{t-1}||^2]} \quad (14)$$

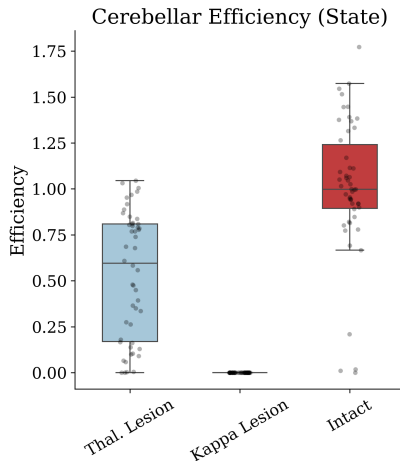
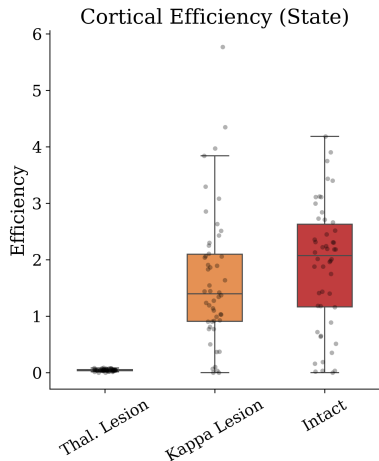
Results: Directional Flow & Raw Encoding (Statistics)

- ▶ **Directional Flow:** Granger causality revealed significantly higher causal flow from Cerebellum $\rightarrow$ Cortex across Value ($\Delta F = 4.70, p = .006$), State ($\Delta F = 3.20, p = .012$), and Volatility ($\Delta F = 9.70, p < .0001$).
- ▶ **Stabilized Efficiency:** The Cortex demonstrated significantly higher stabilized readout efficiency (Value $\Delta = 1.73$, State $\Delta = 0.89$, Volatility $\Delta = 1.41$, all $p < .0001$), acting as a temporal stabilizer against rapid motor errors.
- ▶ **Raw Encoding Capacity:** However, the Cerebellum matched or exceeded the Cortex in raw, unpenalized encoding capacity for Value ($p = .241$) and Volatility ($p = .768$), and was significantly higher for State ($p < .0001$).

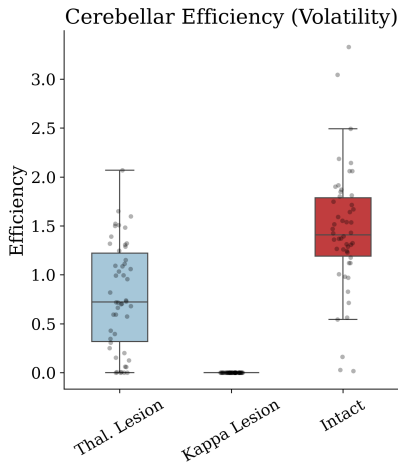
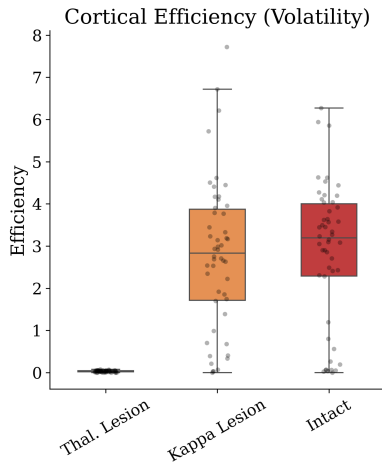
Results: Synthetic Ablations & Action Encoding (Value)



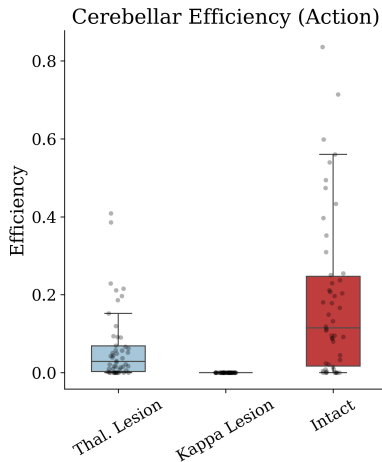
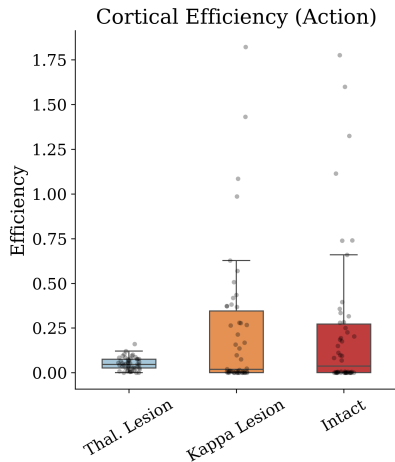
Results: Synthetic Ablations & Action Encoding (State)



Results: Synthetic Ablations & Action Encoding (Volatility)



Results: Synthetic Ablations & Action Encoding (Action)



Results: Synthetic Ablations & Action Encoding (Statistics)

- ▶ **Cortical Double Dissociation:** Severing thalamic communication (Lesion_{Thal}) significantly degraded Cortical efficiency across all domains (Value $\Delta = -3.13$, State $\Delta = -1.85$, Volatility $\Delta = -2.82$, Action $\Delta = -0.186$, all $p < .01$). Disabling climbing fiber plasticity (Lesion_{κ}) produced no statistically significant reduction.
- ▶ **Cerebellar Double Dissociation:** The dependency was inverted. Disabling plasticity (Lesion_{κ}) significantly collapsed Cerebellar efficiency (Value $\Delta = -1.49$, State $\Delta = -1.01$, Volatility $\Delta = -1.45$, Action $\Delta = -0.187$, all $p < .0001$). While Lesion_{Thal} also reduced efficiency, it was significantly less severe than the direct degradation caused by Lesion_{κ} (Value $\Delta = 0.675$, $p < .0001$).

Methods: Quantifying Policy Perseverance (Math & Intuition)

- ▶ **Heuristic Definition:** We defined a Win-Stay/Lose-Shift policy event. If a participant received a reward and repeated the action, it is a "Win-Stay" (w). If they missed and switched, it is a "Lose-Shift" (l).
- ▶ **Bayesian Update:** Instead of a static ratio, we modeled their continuous policy state on a trial-by-trial basis using a conjugate Beta posterior:

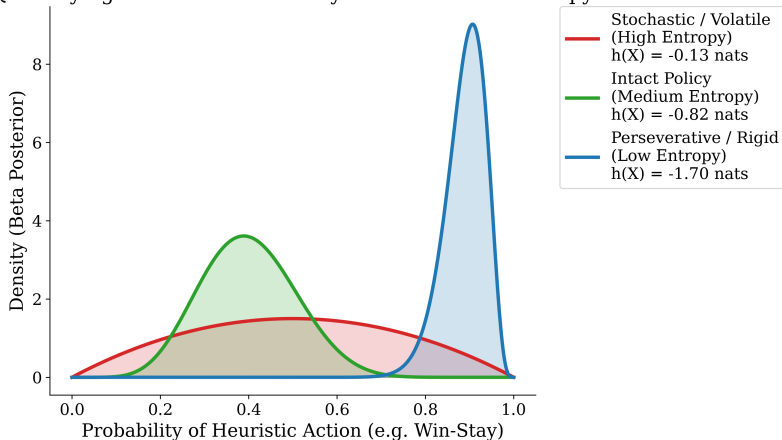
$$X \sim \text{Beta}(\alpha_0 + w, \beta_0 + l) \quad (15)$$

where $\alpha_0 = \beta_0 = 1$ acts as a uniform prior.

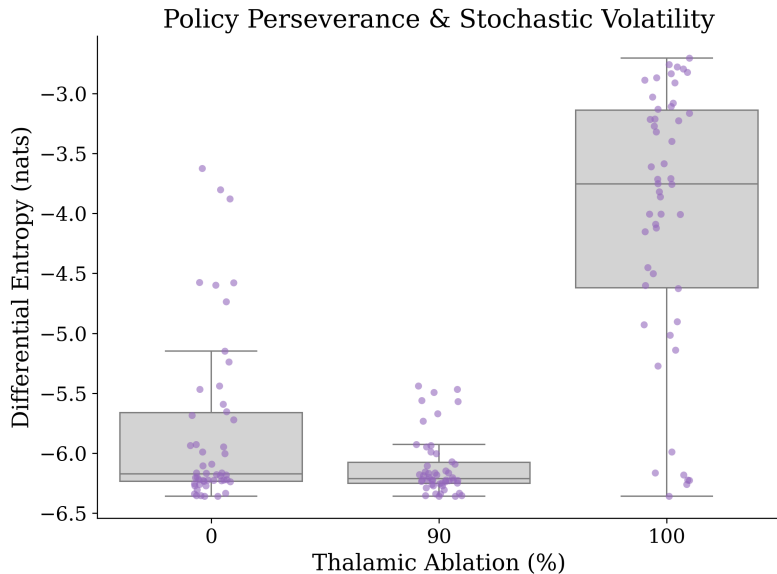
- ▶ **Differential Entropy ($h(X)$):** We computed the differential entropy of this continuous Beta distribution. **Intuition:** It measures the "width" or uncertainty of the policy. If the network is trapped in a rigid heuristic (e.g., 100% Win-Stay), the distribution spikes and entropy collapses.

Methods: Quantifying Policy Perseverance (Graphical Example)

Quantifying Behavioral Flexibility via Differential Entropy



Results: Policy Perseverance & Entropy



Results: Policy Perseverance & Entropy (Statistics)

- ▶ **Heuristic Perseverance:** At 90% partial ablation, differential entropy dropped significantly (Estimate = -0.298, $p = .007$), trapping M_{cc} in a rigid perseverative strategy mimicking clinical symptoms.
- ▶ **Pathological Volatility:** At 100% complete destruction, the network shattered into a highly volatile stochastic state (Estimate = +1.783, $p < .0001$).
- ▶ **Trial-by-Trial Online Readout:** Both ablated extremes suffered massive online Choice NLL predictive interference compared to the Intact network (90% Ablation $t(98) = -7.810, p < .0001$; 100% Ablation $t(98) = -7.959, p < .0001$).

Discussion

- ▶ **The Core Finding:** The cortico-cerebellar loop is strictly required to buffer historical action trajectories and overcome behavioral perseveration in non-Markovian environments.
- ▶ **A Topological Necessity:** Bayesian estimation confirms that unconstrained recurrent states overfit; tight topological bounds (like the granular expansion and Purkinje compression) are required for reliable, out-of-sample decision making.
- ▶ **Future Directions:** Future models should leverage direct physiological data to inform Bayesian priors, grounding the synthetic hyperparameters ($\alpha_c, \kappa, \beta_{thal}$) directly in in-vivo cerebellar firing rates and temporal dynamics.